Data Communication & Computer Networks — Topic 5: Transmission Medium

Practice Test Booklet — 60 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A — Signals & Performance (Encoding, Noise, Bandwidth, Data Rate)

Q1. 'Manchester Encoding' and 'Differential Manchester Encoding' are signal encoding techniques used at which layer?

'Manchester Encoding' और 'Differential Manchester Encoding' signal encoding techniques किस layer पर उपयोग होती हैं?

(A) Physical Layer **(B)** Data Link Layer
(C) Network Layer **(D)** Presentation Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q2. A channel has high bandwidth but suffers from high noise. Which encoding scheme gives better error resistance?

एक channel की bandwidth अधिक है लेकिन noise बहुत है। कौन-सी encoding scheme बेहतर error resistance देगी?

(A) NRZ **(B)** Manchester
(C) AMI **(D)** ASK

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q3. Reduction in signal strength over distance is called:

दूरी के साथ signal strength में कमी को क्या कहते हैं?

(A) Distortion **(B)** Noise
(C) Attenuation **(D)** Interference

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) · also CN Topic Test-04] · Easy

Q4. In a copper cable, attenuation increases with frequency. Which change will REDUCE attenuation?

Copper cable में attenuation frequency के साथ बढ़ता है। कौन-सा बदलाव attenuation घटाएगा?

(A) Increase cable length **(B)** Use lower-category UTP
(C) Use thicker copper core **(D)** Increase signal frequency

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q5. Which noise is caused by electromagnetic coupling between nearby cables?

पास की cables के बीच electromagnetic coupling से कौन-सा noise होता है?

(A) Thermal noise **(B)** Crosstalk
(C) Impulse noise **(D)** Quantization noise

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q6. Which impairment cannot be completely eliminated but only minimised?

कौन-सा impairment पूरी तरह समाप्त नहीं किया जा सकता, केवल कम किया जा सकता है?

(A) Attenuation **(B)** Distortion
(C) Noise **(D)** Delay

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q7. Which parameter of an analog signal represents the number of cycles per second?

Analog signal का कौन-सा parameter प्रति सेकंड cycles की संख्या दर्शाता है?

(A) Amplitude **(B)** Phase
(C) Frequency **(D)** Wavelength

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q8. Which of the following Ethernet encodings has the highest redundancy?

निम्न में से कौन-सी Ethernet encoding की redundancy सबसे अधिक है?

(A) 1B2B **(B)** 4B5B
(C) 5B6B **(D)** 8B10B

[Source: Unit Test-4 Solution (OCR), p9 (options recovered from source)] · Medium

Q9. Which encoding technique is used to transmit signals in Gigabit Ethernet over fiber optic medium?

Fiber optic medium पर Gigabit Ethernet में signal भेजने के लिए कौन-सी encoding technique उपयोग होती है?

(A) Differential Manchester encoding **(B)** Non-Return-to-Zero (NRZ)
(C) 4B/5B **(D)** 8B/10B

[Source: Unit Test-4 Solution (OCR), p30 (options recovered from source)] · Medium

Q10. _____ signals are discrete and are generated by digital modulation.

_____ signals discrete होते हैं और digital modulation से बनते हैं।

(A) Analog **(B)** Digital
(C) Noise **(D)** Bandwidth

[Source: Random Test- Rajasthan PYQ (OCR), p5 — Q108 (options recovered)] · Medium

Q11. Quantizing noise occurs in:

Quantizing noise किसमें होती है?

(A) PCM (Pulse Code Modulation) **(B)** TDM
(C) FDM **(D)** PPM

[Source: Random Test- Rajasthan PYQ (OCR), p10 — Q39 (options recovered)] · Medium

Q12. The effective bandwidth of a communication channel is primarily limited by:

Communication channel की effective bandwidth मुख्य रूप से किससे सीमित होती है?

(A) Protocol overhead **(B)** Noise and attenuation
(C) File size **(D)** MAC address filtering

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q13. What technique allows multiple signals to share the same communication channel?

कौन-सी technique कई signals को एक ही communication channel share करने देती है?

(A) Multiplexing **(B)** Phase modulation
(C) Analog modulation **(D)** Digital modulation

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q14. Communication using a modulated carrier to apply a data signal to a transmission medium in analog form is:

Modulated carrier का उपयोग करके data signal को analog रूप में transmission medium पर डालना क्या कहलाता है?

(A) Broadband networking **(B)** Baseband networking
(C) Modem **(D)** Local area network

[Source: IBPS RRB PO Computer Awareness PYQ (via Meritnotes)] · Medium

Q15. The amount of data transmitted in a fixed amount of time is called:

निश्चित समय में भेजे गए data की मात्रा को क्या कहते हैं?

(A) Refresh rate **(B)** Resolution
(C) Bandwidth **(D)** Frequency

[Source: Unit Test-4 Solution (OCR), p23] · Medium

Q16. _____ signals are discrete and are generated by digital modulation.

_____ signals discrete होते हैं और digital modulation से बनते हैं.

(A) Digital **(B)** Noise
(C) Bandwidth **(D)** Analog

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

Q17. Communication using a modulated carrier to apply a data signal to a transmission medium in analog form is called:

Modulated carrier का उपयोग करके data signal को analog रूप में transmission medium पर भेजना कहलाता है:

(A) Broadband communication **(B)** Baseband communication
(C) Modem **(D)** Local area network

[Source: IBPS RRB PO Computer Awareness PYQ (via meritnotes)] · Medium

Q18. Identify the odd one out: Optical fibre, Twisted pair wire, Microwaves, Hertz

असंगत वस्तु पहचानें: Optical fibre, Twisted pair wire, Microwaves, Hertz

(A) Optical fibre **(B)** Twisted pair wire
(C) Microwaves **(D)** Hertz

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

Q19. Which encoding has a transition in the middle of every bit and is self-synchronizing?

कौन-सी encoding में हर bit के बीच transition होता है और वह self-synchronizing है?

(A) NRZ-L **(B)** Manchester
(C) Unipolar **(D)** Analog

[Source: Practice — original (textbook-based)] · Medium

Q20. Attenuation of a signal means:

Signal का attenuation अर्थ है:

(A) Its strength decreases with distance / दूरी के साथ शक्ति घटना **(B)** Its frequency increases / आवृत्ति बढ़ना
(C) It becomes digital / वह digital होना **(D)** It gets an error / उसमें त्रुटि आना

[Source: Practice — original (textbook-based)] · Medium

Q21. Crosstalk is a type of:

Crosstalk किसका प्रकार है?

(A) Noise **(B)** Encoding
(C) Protocol **(D)** Topology

[Source: Practice — original (textbook-based)] · Medium

Q22. The primary use of scramblers in line coding schemes is to:

Line coding schemes में scramblers का मुख्य उपयोग क्या है?

(A) Make the data look random **(B)** Compress transmission data
(C) Increase signal bandwidth **(D)** Decrease signal bandwidth

[Source: RPSC Programmer 2024 Paper-1 (official key)] · Hard

Q23. The period of a signal is 100 ms. What is its frequency in kilohertz?

एक signal की period 100 ms है। इसकी frequency kilohertz में कितनी होगी?

(A) 10^{-1} kHz **(B)** 10^{-2} kHz
(C) 10^{-3} kHz **(D)** 10^{-4} kHz

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

Q24. If the channel bandwidth is limited to 6 kHz and the signal-to-noise ratio is 16, what is the channel capacity?

यदि channel bandwidth 6 kHz सीमित है और signal-to-noise ratio 16 है, तो channel capacity कितनी होगी?

(A) 15.15 kbps **(B)** 24.74 kbps
(C) 30.12 kbps **(D)** 52.18 kbps

[Source: Random Test- Rajasthan PYQ (OCR), p30 — Q45 (options recovered)] · Hard

Q25. Given a bit rate of b bits/sec, the time required to send 16 bits is:

यदि bit rate b bits/sec है, तो 16 bits भेजने में कितना समय लगेगा?

(A) 16 × b sec **(B)** 16/b sec
(C) 16b sec **(D)** b/16 sec

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

Q26. An analog signal carries 4 bits per signal unit. If 1000 signal units are sent per second, what are the baud rate and bit rate?

एक analog signal प्रति signal unit 4 bits ले जाती है। यदि प्रति सेकंड 1000 signal units भेजे जाएँ, तो baud rate और bit rate क्या होंगे?

(A) 1000 bauds/sec, 4000 bps **(B)** 1000 bauds/sec, 500 bps
(C) 4000 bauds/sec, 1000 bps **(D)** 2000 bauds/sec, 1000 bps

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

Q27. Consider a noiseless channel with a bandwidth of 5000 Hz transmitting a signal with two signal levels. The maximum bit rate is:

एक noiseless channel जिसकी bandwidth 5000 Hz है, दो signal levels के साथ transmission कर रहा है. अधिकतम bit rate होगी:

(A) 2500 bps **(B)** 10000 bps
(C) 5000 bps **(D)** 20000 bps

[Source: UGC NET Computer Science Jan 2025 (via samagracs solved paper)] · Hard

Q28. Which of the following is NOT a type of broadband internet connection?

निम्न में से कौन-सा broadband internet connection का प्रकार नहीं है?

(A) Cable **(B)** DSL
(C) Satellite **(D)** Dial-up

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Hard

Q29. According to Nyquist, the maximum bit rate of a noiseless channel with bandwidth 3000 Hz and 4 signal levels is:

Nyquist के अनुसार 3000 Hz bandwidth, 4 signal levels वाले noiseless channel की अधिकतम bit rate है:

(A) 6000 bps **(B)** 12000 bps
(C) 24000 bps **(D)** 3000 bps

[Source: Practice — original (textbook-based)] · Hard

Q30. A signal-to-noise ratio of 1000 corresponds to how many dB?

Signal-to-noise ratio 1000 कितने dB है?

(A) 3 dB **(B)** 10 dB
(C) 30 dB **(D)** 100 dB

[Source: Practice — original (textbook-based)] · Hard

Section B — Guided Media (UTP, Coaxial, Fiber Optic)

Q31. Optical fiber uses reflection to guide light through a channel. At total internal reflection, the angle of incidence is _____ the critical angle.

Optical fiber प्रकाश को channel में भेजने के लिए reflection उपयोग करती है। Total internal reflection में angle of incidence, critical angle से _____ होता है।

(A) Equal to **(B)** Less than
(C) Greater than **(D)** Less than or equal to

[Source: Repo: CN Topic Test-04 (UNIT TEST-4) — recovered from BCI T8 Q33 options] · Easy

Q32. A network uses coaxial cable but suffers from signal reflection. What is the most likely cause?

एक network coaxial cable उपयोग करता है लेकिन signal reflection से पीड़ित है। सबसे संभावित कारण क्या है?

(A) EMI **(B)** Incorrect termination resistance
(C) Crosstalk **(D)** Low bandwidth

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q33. Which transmission medium is primarily used in modern internet backbone networks?

आधुनिक internet backbone networks में मुख्य रूप से कौन-सा transmission medium उपयोग होता है?

(A) Twisted pair cable **(B)** Coaxial cable
(C) Single mode fiber optic cable **(D)** Microwave links

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q34. Which cable has the maximum bandwidth?

किस cable की bandwidth अधिकतम होती है?

(A) Coaxial **(B)** Twisted pair
(C) Optical fiber **(D)** USB cable

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q35. Assertion (A): Fiber optic cable is immune to electromagnetic interference (EMI). Reason (R): It transmits data in the form of light pulses rather than electrical signals. Which option is correct?

Assertion (A): Fiber optic cable electromagnetic interference (EMI) से मुक्त है। Reason (R): यह data electrical signals के बजाय light pulses के रूप में भेजती है। कौन-सा विकल्प सही है?

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A) **(B)** Both (A) and (R) are true, but (R) is NOT the correct explanation
(C) (A) is true, but (R) is false **(D)** (A) is false, but (R) is true

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) — Assertion-Reason] · Medium

Q36. Which statements are TRUE for fiber optic cables? I. Immune to tapping II. High bandwidth III. Requires electrical grounding IV. Works using total internal reflection

Fiber optic cables के लिए कौन-से कथन सही हैं? I. Tapping से मुक्त II. उच्च bandwidth III. Electrical grounding चाहिए IV. Total internal reflection से काम करती है

(A) I, II and IV **(B)** II and III
(C) I, III and IV **(D)** I and II

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) — Q33] · Medium

Q37. A company needs long-distance communication (5-10 km), high data rate and immunity from electromagnetic interference. Which medium is best?

एक कंपनी को लंबी दूरी (5-10 km) संचार, उच्च data rate और EMI से मुक्ति चाहिए। कौन-सा medium सर्वोत्तम है?

(A) UTP **(B)** Coaxial cable
(C) Single-mode fiber **(D)** Multi-mode fiber

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q38. A UTP cable used in Gigabit Ethernet shows high delay skew. What is the cause?

Gigabit Ethernet में उपयोग UTP cable में delay skew अधिक है। कारण क्या है?

(A) Different twist rates in the cable pairs **(B)** Excessive EMI
(C) Fiber splice error **(D)** Wrong wireless channel

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q39. Which of the following best explains why fiber optic cables have a lower bit error rate than copper cables?

Fiber optic cables की copper cables की तुलना में कम bit error rate की सबसे अच्छी व्याख्या कौन-सी है?

(A) Higher signal power **(B)** Immunity to electromagnetic interference
(C) Simpler modulation techniques **(D)** Larger physical size

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q40. Which transmission medium works on the principle of total internal reflection?

कौन-सा transmission medium total internal reflection के सिद्धांत पर काम करता है?

(A) Optical fiber cable **(B)** Shielded twisted pair
(C) Unshielded twisted pair **(D)** Coaxial cable

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q41. Crosstalk in twisted pair cables is typically minimised by:

Twisted pair cables में crosstalk आमतौर पर किससे कम होता है?

(A) Increasing the power level **(B)** Increasing the twist rate
(C) Adding shielding to the core **(D)** Using longer cables

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q42. Which connector is commonly used for coaxial cable in physical networks?

Physical networks में coaxial cable के लिए सामान्यतः कौन-सा connector उपयोग होता है?

(A) RJ-45 **(B)** USB-C
(C) BNC **(D)** HDMI

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q43. Which transmission medium is best for a network exposed to heavy electrical interference?

भारी electrical interference वाले network के लिए कौन-सा transmission medium सर्वोत्तम है?

(A) Unshielded twisted pair **(B)** Optical fiber
(C) Coaxial cable **(D)** Microwave

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

Q44. Which of the following CANNOT be used as a medium for 802.3 Ethernet?

निम्न में से कौन-सा 802.3 Ethernet के लिए medium उपयोग नहीं किया जा सकता?

(A) Thin coaxial cable **(B)** Twisted pair cable
(C) Microwave link **(D)** Fiber optic cable

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

Q45. Which fiber type is used for long-distance communication?

लंबी दूरी के संचार के लिए कौन-सी fiber प्रयोग होती है?

(A) Multimode fiber **(B)** Single-mode fiber
(C) Coaxial cable **(D)** UTP

[Source: Practice — original (textbook-based)] · Medium

Q46. UTP cables are normally terminated with which connector?

UTP cables सामान्यतः किस connector से जुड़ती हैं?

(A) BNC **(B)** RJ-45
(C) SC **(D)** USB

[Source: Practice — original (textbook-based)] · Medium

Q47. Cable TV networks commonly use which cable impedance?

Cable TV networks सामान्यतः किस impedance की cable प्रयोग करते हैं?

(A) 50 ohm **(B)** 75 ohm
(C) 100 ohm **(D)** 600 ohm

[Source: Practice — original (textbook-based)] · Medium

Q48. For an office LAN needing low cost and easy installation up to 100 m, the best choice is:

100 m तक, कम लागत और आसान installation वाले office LAN के लिए सबसे अच्छा विकल्प:

(A) Fiber optic **(B)** UTP
(C) Coaxial **(D)** Satellite

[Source: Practice — original (textbook-based)] · Medium

Q49. What is the typical maximum length of a UTP cable in Ethernet before signal degradation?

Signal degradation से पहले Ethernet में UTP cable की सामान्य अधिकतम लंबाई कितनी होती है?

(A) 50 meters **(B)** 100 meters
(C) 150 meters **(D)** 200 meters

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

Q50. Which of the following is NOT a type of transmission medium?

निम्न में से कौन-सा transmission medium का प्रकार नहीं है?

(A) Coaxial cable **(B)** Fiber optics
(C) NIC **(D)** Twisted pair

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

Section C — Unguided Media (Wireless)

Q51. Which of the following is a wireless networking technology?

निम्न में से कौन-सी wireless networking technology है?

(A) Ethernet **(B)** Wi-Fi
(C) Fiber Optic **(D)** DSL

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q52. Which wireless standard provides the highest data rate?

कौन-सा wireless standard सर्वोच्च data rate देता है?

(A) 802.11b **(B)** 802.11g
(C) 802.11n **(D)** 802.11ac

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q53. Which type of wireless transmission requires line-of-sight?

किस प्रकार की wireless transmission के लिए line-of-sight आवश्यक है?

(A) Wi-Fi **(B)** Infrared
(C) Radio **(D)** Bluetooth

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q54. Which transmission medium is used in Bluetooth technology?

Bluetooth technology में कौन-सा transmission medium उपयोग होता है?

(A) Radio links **(B)** Microwave links
(C) VSAT communication **(D)** Fiber optic

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q55. Which physical-layer technique allows multiple users to share the same bandwidth simultaneously?

कौन-सी physical-layer technique कई users को एक ही bandwidth साथ-साथ share करने देती है?

(A) TDMA **(B)** FDMA
(C) CDMA **(D)** ARQ

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q56. Microwave communication requires:

Microwave communication के लिए आवश्यक है:

(A) Line of sight **(B)** Wires / तार
(C) Fiber cable **(D)** Underground cable / भूमिगत cable

[Source: Practice — original (textbook-based)] · Medium

Q57. Bluetooth is designed for which type of network?

Bluetooth किस प्रकार के network के लिए बना है?

(A) WAN **(B)** PAN
(C) SAN **(D)** CAN

[Source: Practice — original (textbook-based)] · Medium

Q58. Which of the following statements is NOT true about microwaves?

Microwaves के बारे में निम्न में से कौन-सा कथन सही नहीं है?

(A) They are electromagnetic waves with frequencies from 300 GHz to 400 THz **(B)** Propagation is line-of-sight
(C) Very high-frequency waves cannot penetrate walls
(D) Use of certain portions of the band requires permission from authorities

[Source: Repo: CN Topic Test-04 (UNIT TEST-4) — options cleaned from source] · Hard

Q59. Select the FALSE statement about CDMA technology: I. CDMA optimises the use of available bandwidth as it transmits over the entire frequency range. II. Each user in a CDMA system uses a different code to modulate their signal. III. CDMA offers less security compared to GSM technology. IV. A SIM card is always required for a CDMA device to work.

CDMA technology के बारे में गलत कथन चुनिए: I. CDMA उपलब्ध bandwidth के उपयोग को अनुकूलित करता है क्योंकि यह पूरी frequency range पर transmit करता है। II. CDMA system में हर user अपने signal को modulate करने के लिए अलग code उपयोग करता है। III. CDMA, GSM की तुलना में कम security देता है। IV. CDMA device के काम करने के लिए SIM card हमेशा आवश्यक है।

(A) Only (I) **(B)** Only (II)
(C) (I) and (II) **(D)** (III) and (IV)

[Source: Random Test- Rajasthan PYQ (OCR), p11 — Q83 (statements recovered from screenshot)] · Hard

Q60. A geostationary satellite orbits approximately at a height of:

Geostationary satellite लगभग किस ऊँचाई पर चक्कर लगाता है?

(A) 36,000 km **(B)** 500 km
(C) 1,000 km **(D)** 100 km

[Source: Practice — original (textbook-based)] · Hard

Answer Key — sirf letters (no solutions)

Q1: A **Q2:** B **Q3:** C **Q4:** C **Q5:** B **Q6:** C **Q7:** C **Q8:** A **Q9:** D **Q10:** B
Q11: A **Q12:** B **Q13:** A **Q14:** A **Q15:** C **Q16:** A **Q17:** A **Q18:** D **Q19:** B **Q20:** A
Q21: A **Q22:** A **Q23:** B **Q24:** B **Q25:** B **Q26:** A **Q27:** B **Q28:** D **Q29:** B **Q30:** C
Q31: C **Q32:** B **Q33:** C **Q34:** C **Q35:** A **Q36:** A **Q37:** C **Q38:** A **Q39:** B **Q40:** A
Q41: B **Q42:** C **Q43:** B **Q44:** C **Q45:** B **Q46:** B **Q47:** B **Q48:** B **Q49:** B **Q50:** C
Q51: B **Q52:** D **Q53:** B **Q54:** A **Q55:** C **Q56:** A **Q57:** B **Q58:** A **Q59:** D **Q60:** A